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EDUCATION

Nanyang Technological University, SINGAPORE Aug 2023 - Dec 2026
Bachelor of Engineering (Hons), Computer Engineering, GPA: 4.66 / 5.0
Relevant Coursework: Computer Vision, Machine Learning, Embedded Programming (FreeRTOS, embedded Linux)

Peking University, CHINA (Exchange) Jan 2025 - Jun 2025

Singapore Polytechnic, Diploma in Information Technology, GPA: 3.70 / 4.0 Apr 2018 - Apr 2021

WORK EXPERIENCE

Black Sesame Technologies May 2026 - Oct 2026
Embodied AI Intern
- Built a demonstration generator in Isaac Lab with automated checkpoint evaluation in Isaac Sim; fully fine-tuned Physical Intelligence's π0.5 (pi0.5, openpi), reaching 100% success on simulated pick-and-place.
- Deployed π0.5 on a real Franka Research 3 (FR3) via a ROS 2 bridge at 10 Hz; built a teach-and-repeat recorder (hand-guide once, robot replays) that captured occlusion-free demonstrations.
- Architected an agentic planner around a frozen π0.5 policy, integrating SAM 3 open-vocabulary segmentation, analytic kinematics, and visual calibration to autonomously sequence multi-step tasks such as cube stacking on the real arm.
- Implemented real-time single-camera MediaPipe teleoperation of an Inspire RH56 dexterous hand; built a custom CAN driver with grip-force limiting to actuate it.

ST Engineering Land Systems Jul 2025 - Dec 2025
Intern (Embedded Systems) · Awarded the Intern @ ST Engineering Award 2025
- Developed a scalable diagnostic and calibration framework for wheeled platforms with Hybrid Electric Drive architecture, designed for reuse across multiple vehicle variants.
- Built a WPF diagnostic interface (C#/.NET) with real-time CAN bus communication and data visualisation.
- Implemented Structured Text (CODESYS) firmware on the Mobility Control Unit (STW ESX-3CM) for persistent fault logging and EEPROM data management, including wear levelling and CRC-based integrity checks.

Wistech Consulting Pte Ltd Feb 2021 - May 2021
Software Developer
- Built and maintained multi-client Shopify storefronts and themes.
- Developed a product aggregation web app with ASP.NET and SQL Server.

PROJECT WORK

Final Year Project Dec 2025 - Present
Robot Foundation Models (Franka Emika Panda)
- Reimplemented Instant Policy (graph-diffusion manipulation policy) in PyTorch for in-context imitation learning, encoding raw point clouds into scene graphs to generate SE(3) end-effector trajectories from reference demonstrations without task-specific retraining.
- Built a closed-loop task controller in which a Qwen3-VL agent chains short-horizon sub-tasks into long-horizon tasks, integrating a learned demonstration selector (which demos) and a learned boundary monitor (when to switch), with autonomous error recovery/retries to eliminate human-in-the-loop intervention.

Multidisciplinary Design Project Jan 2026 - May 2026
STM32 Sensor-Driven Mobile Robot
- Engineered bare-metal C/C++ firmware for an STM32 mobile robot that communicates with Raspberry Pi over UART, configuring hardware peripherals (GPIO, ADC, timers) and optimising sensor polling rates for reliable signal timing.
- Synchronised wheel speeds with encoders and auto-corrected heading with a complementary-filtered IMU for straight runs and set-angle turns.
- Integrated a Sharp IR sensor for wall detection and following, and an HC-SR04 ultrasonic sensor for distance sensing, using both as range thresholds in the closed-loop motion control.

SKILLS

Robot Learning:	Vision-Language-Action (VLA), diffusion policies, imitation learning, PyTorch, SAM 3, MediaPipe
Robotics:	ROS 2, libfranka, Isaac Sim / Lab, Franka robot arm, Inspire dexterous hand, Orbbec RGB-D
Embedded:	STM32, FreeRTOS, CAN, UART, SPI / I2C, PREEMPT_RT Linux, GDB, FPGA
Languages & Tools:	Python, C/C++, C#, Verilog, Assembly (ARM Cortex-M, x86-64), ASP.NET / WPF, Git / GitHub